

Govind Mittal

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Professional Summary

Data Scientist / AI Engineer (AI / LLM / Agentic Systems) with 3+ years deploying **LLM-powered, agentic, and RAG solutions** for public safety, finance, and operational decision-support. Expert in **LangGraph, MCP, LangSmith**, knowledge graphs, and **explainable AI** for **high-stakes environments**. Proven track record building **production-grade agentic systems** with safety guardrails and real-time operational workflows.

Work Experience

Eli Lilly — Senior Scientist, AI Tech at Lilly

Apr 2025 – Nov 2025 — Bengaluru, India

- **CFO Notes System (LLM Agents for Finance Ops):** Architected **multi-agent RAG LangGraph + FastAPI** platform providing **real-time decision-support** for financial variance reporting across 20 products/18 markets (1 day → 15 min). Engineered **ambient, long-running agent** with **safety guardrails and explainability features**; became flagship tool for **Lilly USA Finance** supporting high-stakes financial operations.
- **Nexify (Data Science Automation Platform):** Built async agentic pipelines with **LangGraph + Human-in-the-Loop + time-travel + subagents** controls, shortening question-to-insight cycle from weeks to minutes. Designed **comprehensive evaluation suites via LangSmith** measuring accuracy, coherence, bias, and **operational safety metrics** for non-deterministic LLM outputs in **production environments**.
- **VizSnap MCP Server:** Developed **MCP-based agentic AI visualization system** integrating LLMs with **external data APIs**, enabling dynamic dashboards (ECharts) and multi-chart stitching. **Deployed organization-wide** with 900+ active users processing 10K+ queries/month. Finetuned Qwen 2.5-VL using RL via Agent Lightning.
- **ARTIE Platform (Artificial Teams for Rapid Insights):** Unified CFO Notes, Nexify, and VizSnap as modular agentic blocks (React, LLM Tool Call, Validator, Reviewer) enabling **automated agentic operational workflows like LangFlow**; scaled from 300 → 900+ users across enterprise.

Citigroup — Senior Data Scientist

Jul 2022 – Mar 2025 — Bengaluru, India

- **Project Luna (LLM + Knowledge Graph IVR):** Built hybrid intent-detection and **real-time decision-support system** using **embeddings + Neo4j knowledge-graph reasoning**; increased IVR containment 12%, saving \$2.4M/month. **Deployed to production** processing 50K+ daily interactions with 94% accuracy in mission-critical customer service operations.
- **Compliance QC & Governance:** Developed **multi-agent RAG framework** for regulatory compliance and **model risk management** with **audit trails and explainability**. Reduced analyst time 52%, QC misses 6% while ensuring regulatory oversight.
- **Project Chronos:** Developed forecasting models using **hierarchical time-series + Transformer ensembles**, improving accuracy from 81% → 91.8% for front office operational capacity planning and resource allocation.

Skills & Tools

Core AI & Agentic Frameworks: LLMs (GPT, Claude, Gemini, Llama), Agentic-AI, RAG Pipelines, Prompt-Engineering, Fine-tuning (LoRA, QLoRA) LangGraph, LangChain, MCP, LangSmith, DSPy, Hugging Face, Unsloth, Llama, Mistral

AI Safety & Evaluation: Model Safety, Explainable AI (SHAP, LIME), Bias Detection, Evaluation Frameworks, Human-in-the-Loop Systems, Safety Metrics for Non-Deterministic Outputs

Knowledge Graphs & Vector DBs: Neo4j, AWS Neptune, Pinecone, Weaviate, Chroma, FAISS, Elasticsearch, Semantic Search, Entity Linking, Relationship Modeling

ML & Deep Learning: PyTorch, TensorFlow, Keras, Reinforcement Learning, Forecasting, Anomaly Detection

Languages & APIs: Python, R, SQL, FastAPI, Flask, REST/GraphQL, API Development, Microservices

DevOps & Cloud: Docker, Kubernetes, Git, CI/CD, MLOps, GCP, AWS, Azure, GPU Clusters

Data Engineering: Snowflake, PostgreSQL, MongoDB, PySpark, Pandas, NumPy, Big Data Processing

Visualization: Apache ECharts, Tableau, Power BI, Plotly, Streamlit, Matplotlib, Seaborn, Interactive Dashboards

Projects

- **Daedalus – The Deep Researcher:** Multi-agent system for parallel web/file exploration using LangGraph deep agents with dynamic sub-agent spawning, memory management, and automated synthesis workflows.
- **Mechanistic Interpretability Replication:** Reproduced Anthropic's *attribution-graph approach* for visualizing transformer neuron pathways and interpretability; implemented circuit discovery techniques for trustworthy AI development.
- **AI Ghalib: Finetuned Gemma 2 9b** via unsloth on Shayaris of Ghalib scraped from rekhta using **QLORA** to be an ai shayar and generate shayaris.

Education

MSc Economics – Gokhale Institute of Politics & Economics (Pune, India)

2020 – 2022

Awards: Batch Prize – Highest Scores in Econometrics, Business Analytics, Linear Economics

BSc Economics & Statistics – N.M.I.M.S University (Mumbai, India)

2015 – 2018

Awards: Best Paper Presentation & Budding Economist Awards